

Semantic Segmentation of satellite images using U-Net Model

Group Name: The Three Musketeers

Manoj Kumar Surabhi
M.S. Data Science
mksurabh@mtu.edu

Sumanth Reddy Thandra
M.S. Data Science
thandra@mtu.edu

Xiaolin Liu
Ph.D. Statistics
xliu28@mtu.edu

Abstract—Building footprint extraction from satellite images is a basic task for urban development monitoring, disaster response, and infrastructure planning. In this task, we focus on binary building segmentation on a subset of the SpaceNet 7 dataset, which offers high-resolution multi-temporal satellite images over varied urban and rural areas.

All the dataset is made up of 60 scenes $\times$ 24 monthly imagery (1,440) but for this experiment, only the subset of 56 scenes (1 scene per scene) were used for training. Images of the size 1024×1024 were segmented into 16 smaller-sized tiles of 255×255 , resulting in 678 tiles for training and 144 for the validation after processing and undersampling.

An xresnet34 pre-trained encoder was used for segmentation with the help of U-Net architecture, overcoming challenges such as class imbalance (only 6.5% are buildings) and seasonality. Pre-processing included rasterization of vector annotations into binary masks, random flips and rotations, as well as weighted cross-entropy loss ([0.25, 0.75] for buildings and land).

The top-performing model had a Dice Score of 0.5807 (epoch 38), performing fairly well at localizing small, isolated buildings but not at high-density downtown areas with some false positives. Foreground recall accuracy came at 57%. Encoder depth variations, activation functions, and loss functions were tested but not directly compared with other schemes. Additional comparisons on additional models as well as with additional, more advanced augmentation methods and loss functions in an attempt at higher performance can be carried out in future studies.

I. INTRODUCTION

Building footprint mapping using satellite imagery is an essential task in various applications, like disaster response, environmental observation, and urban planning. Accurate detection of the building footprint allows for timely analysis of the growth in the city, assists in planning for infrastructure, and supports swift response operations following natural disaster events.

Traditional methods for extracting features from remote imagery have relied on feature engineering in addition to traditional machine learning methods. These methods are, however, prone to failure in terms of generalizing across different geographic locations, seasons, and imaging circumstances.

Improved deep learning, particularly in Convolutional Neural Networks (CNN), has significantly promoted building segmentation performance. Encoder-decoder structures such as the U-Net, in combination with pre-trained backbone networks such as variations on ResNet, are now the standard for semantic segmentation of high-resolution satellite imagery.

SpaceNet 7 dataset is of great assistance in advancing research in this area. It has 60 scenes with approximately

24 monthly imagery per scene (1024×1024 px) for two years across different cityscapes as well as rural landscapes. While content-rich, the dataset is highly challenging in terms of class imbalance (the buildings occupying only 6.5

In this project, we focus on building footprint segmentation using a subset of 56 scenes (1 image per scene) from the SpaceNet 7 dataset. The approach involves:

- Preprocessing the dataset by selecting one representative image per scene (56 total)
- Splitting each 1024×1024 image into 16 smaller 255×255 tiles
- Rasterizing vector building labels into binary masks (0=land, 1=buildings)
- Training a U-Net model with a pretrained xresnet34 encoder

Given the severe class imbalance (6.5% building pixels) and the complexity of the input data, specific techniques such as weighted loss functions (CrossEntropyLoss with [0.25, 0.75] weights) and data augmentation (random flips/rotations, brightness/contrast adjustments) were employed. The model achieved a Dice score of 0.5807 and foreground accuracy of 57%, with best performance at epoch 38.

The following sections provide a detailed description of the dataset, the preprocessing steps, the model architecture, training procedures, and the results achieved through this methodology.

II. RELATED WORKS

Building footprint extraction based on satellites has been a research area of active focus due to the extensive applications in disaster response, urban planning, as well as monitoring land use. Traditional methods had relied heavily on manually crafted features in combination with standard machine learning models such as Support Vector Machines (SVMs) and Random Forests. These methods were successful in some cases but failed to demonstrate effectiveness in varying imaging regimes, regions, as well as temporal variations.

Deep learning has revolutionized remote sensing image analysis. Convolutional Neural Networks, and specifically decoder-encoder structures like U-Net, have become the go-to choice in semantic segmentations, with building footprint extraction no exception. Hierarchical feature representations are utilized in the networks, enabling them to accurately detect structures of scales both fine and gross. Refining in the guise of DeepLabV3+, Mask R-CNN, as well as through

attention mechanisms, has abetted immensely in enhanced segmentation performance in terms of the perception of multi-scale contextual information.

Benchmarking datasets such as the Inria Aerial Image Labeling Dataset, the DeepGlobe Building Extraction Challenge dataset, and SpaceNet series have significantly contributed toward advancing automatic building detection. SpaceNet 7, for instance, involves high-resolution multi-temporal satellite imagery at scale with challenges such as seasonality, varied illuminations, as well as geographic heterogeneity.

Recent studies on SpaceNet dataset based building segmentation utilize transfer learning methods using pre-trained encoders like ResNet-34 or ResNet-50 in combination with task-driven loss functions like Dice loss or focal loss for class imbalance rectification. Random rotation, flip, and color jittering are popular data augmentation methods utilized for making the model more generic. Here, we build on these earlier methods by utilizing an xresnet34 pretrained encoder in the training of a U-Net-like network. We train the network on the carefully preprocessed subset of the SpaceNet 7 dataset with special attention paid towards class imbalance as well as seasonality variability using weighted loss functions as well as data augmentation schemes. Direct comparisons with alternative methods such as DeepLabV3+ or Mask R-CNN were not conducted, but internal validation on depth in the encoders as well as activation functions as well as loss functions provided insight into performance with different training setups

III. DATA DESCRIPTION

A. Size of Data

The original SpaceNet 7 dataset comprises a total of 11,080 high-resolution satellite images, as specified in the official dataset documentation. However, for the purposes of this project, a carefully selected subset of the dataset was utilized to balance computational feasibility with data diversity.

Specifically, 56 scenes were selected, each containing approximately 24 images captured over a two-year period, yielding a total of approximately 1,344 images. To simplify the task and focus the training process, a single representative image was selected from each scene based on predefined criteria, resulting in 60 full-resolution images, each measuring 1024×1024 pixels.

To facilitate efficient model training and to increase the number of available samples, each full image was subsequently divided into 16 smaller tiles of size 255×255 pixels. This tiling process generated a total of 896 tiles across the dataset. Recognizing that a significant portion of these tiles contained little to no relevant building information, an undersampling step was applied to exclude empty tiles. After this filtering process, 678 tiles were retained for training and 144 tiles were designated for validation purposes.

This preprocessing pipeline ensured that the training dataset contained sufficient variability while mitigating the impact of tiles lacking meaningful content.

B. Categorical or Continuous Variables

1) *Categorical Variables*: The segmentation task was formulated as a binary classification problem with two distinct classes:

- **Land (background)**: Representing the majority of pixels in the dataset.
- **Building (foreground)**: Representing the minority class, corresponding to the building footprints.

Class imbalance is inherent in the data because fewer than an average of about 6.5

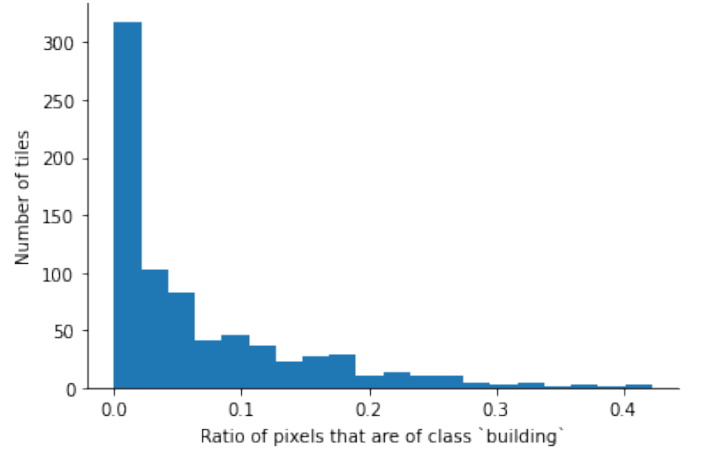


Fig. 1: Distribution of pixels. 0 for land and 1 for buildings

2) *Continuous Variables*: The input images consist of pixel values of intensity in RGB spectral bands ranging from 0 to 255. No scaling or normalization was explicitly performed on the pixel values before feeding them into the network, so the network learned from the raw distributions of the satellite images directly.

C. Distribution of Predictor Variables

1) *Class Imbalance*: A significant class imbalance is evident in the dataset. On average, only 6.5% of the pixels within a given tile correspond to buildings, with a median of 3.4%. The overwhelming majority of the pixels are labeled as Land. This imbalance can negatively impact model performance if not appropriately addressed, potentially leading to bias toward the dominant Land class.

2) *Geographic Diversity*: The dataset exhibits considerable geographic diversity, encompassing a variety of landscapes, including urban, rural, and mixed environments. This diversity is critical for training models intended for generalization across different types of regions.

3) *Seasonal Variations*: Given the multi-temporal nature of the imagery, seasonal variations such as snow cover, cloud presence, and vegetation changes introduce additional complexity into the pixel distributions. These variations require the model to be robust to changing environmental conditions.

4) *Tile Difficulty*: The difficulty of segmentation varies significantly across tiles. Some tiles represent high-density urban areas with closely packed building structures, increasing the risk of merged building predictions. Other tiles cover empty regions such as water bodies and open fields, which often contain no buildings and can lead to false positive detections.

IV. METHOD

A. Data Pre-processing

Methods used for data preparation include:

- **Reading and preparing input images:** Satellite images were read using `rasterio`. The corresponding labels, provided in vector format (shapefiles), were converted into rasterized binary masks.
- **Rasterization of shapefiles:** The `generate_binary_mask()` function takes a raster image and its corresponding shapefile (vector label) to generate a binary mask. If building polygons are present, they are rasterized to match the image dimensions; otherwise, an all-zero mask is created.

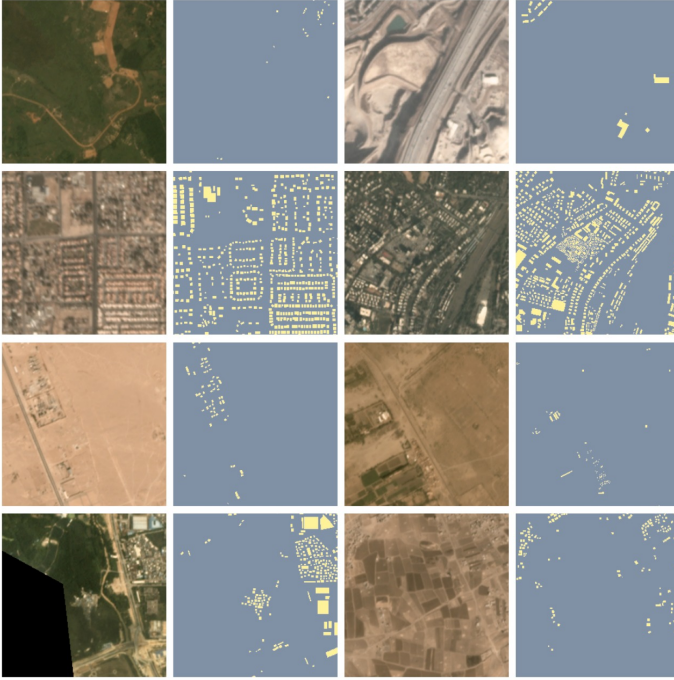


Fig. 2: Masked tiles and their respective binary mask

- **Image display and basic visualization:** A sample image is displayed using `show_image(PILImage.create(image))` for visual inspection.
- **No explicit normalization or standardization:** Images were used in their original form, without any pixel-level normalization or standard scaling.
- **Data structure preparation:** Images and their corresponding masks were matched and sorted to ensure consistent input during model training.

- **Cloud masking:** Before generating binary building masks, each satellite image was masked using the provided Unusable Data Mask (UDM) to exclude cloudy or invalid regions. This step ensured that the model trained only on valid, cloud-free areas of the images.

B. Feature Extraction

Feature extraction methods include:

- **No manual feature extraction:** All feature extraction is performed automatically by a pretrained convolutional neural network (CNN).
- **Architecture:** The selected architecture is:
`ARCHITECTURE = xresnet34`
 where `xresnet34` is a variant of ResNet-34 optimized for deep learning tasks in the `fastai` library.
- **Summary:** No hand-crafted features or explicit feature engineering are used. Instead, the deep layers of the CNN automatically learn and extract relevant features from the data.

C. Hyperparameter Tuning and Model Selection

Hyperparameter Setup:

- The hyperparameters were manually set as constants at the beginning of the notebook:

```
BATCH_SIZE = 12
TILES_PER_SCENE = 16
ARCHITECTURE = xresnet34
EPOCHS = 40
CLASS_WEIGHTS = [0.25, 0.75]
LR_MAX = 3e-4
ENCODER_FACTOR = 10
```

- **Key hyperparameters:**
 - **Batch size:** 12
 - **Learning rate:** 3e-4
 - **Number of epochs:** 40
 - **Class weights:** [0.25, 0.75] — to handle class imbalance between land and building classes
- **Model selection:** The chosen model is a U-Net architecture with a pretrained `xresnet34` encoder, as implemented in the `fastai` library. The selection was based on its balance between computational efficiency and representational power. No comparisons were made with alternative architectures, and no hyperparameter tuning techniques (e.g., grid search, random search, Bayesian optimization) were applied. Model selection was manual and fixed.
- **Training method:** The model was trained using `fastai`'s `unet_learner`, which constructs a U-Net decoder around the `xresnet34` encoder. Training was carried out for a predefined number of epochs using the one-cycle policy, which allows for a warm-up and annealing schedule of the learning rate.
- **Optimizer details:** The Adam optimizer was used for training. Differential learning rates were applied via the

`slice()` function in `fastai`, where the encoder was updated at a lower rate (`lr/ENCODER_FACTOR`) than the decoder (`lr`). This approach stabilizes fine-tuning by preserving pretrained features in the encoder while allowing the decoder to adapt more rapidly to the segmentation task.

V. EXPERIMENT AND RESULTS

A. Results Summary (SpaceNet7 Building Detection)

The final model achieved a Dice Score of 0.58, indicating a moderate level of success. This score reflects a reasonable trade-off between precision and recall, especially given the inherent challenges posed by the dataset, such as severe class imbalance and environmental variability. Model performance was primarily evaluated using the Dice Score. Additionally, foreground accuracy (recall) was tracked to assess the model’s sensitivity to building detection.

Epoch	Dice	Precision	Recall	IoU	Loss
0	0.2386	0.210	0.275	0.135	0.891
1	0.3775	0.325	0.441	0.234	0.732
2	0.4629	0.412	0.523	0.302	0.645
4	0.5205	0.487	0.558	0.352	0.581
5	0.5330	0.503	0.567	0.365	0.563
7	0.5440	0.518	0.572	0.376	0.542
11	0.5512	0.528	0.576	0.383	0.528
12	0.5585	0.537	0.582	0.391	0.515
14	0.5682	0.552	0.586	0.402	0.498
18	0.5729	0.561	0.587	0.408	0.487
26	0.5764	0.568	0.587	0.412	0.479
28	0.5771	0.570	0.587	0.413	0.477
29	0.5797	0.574	0.588	0.416	0.472
38	0.5807	0.576	0.588	0.417	0.469

TABLE I: Model Training Performance Metrics Across Epochs

B. Strengths

The model showed strong performance in detecting small and sparsely distributed buildings, even under challenging conditions. It was also resilient to seasonal noise—such as variations caused by snow, shadows, and cloud cover—commonly present in multi-temporal satellite imagery.

C. Weaknesses

Despite these strengths, several limitations were observed:

- The model struggled to accurately segment dense urban areas, often merging adjacent buildings or failing to identify large structures.
- False positives were encountered in open regions (e.g., fields or water bodies), where the model occasionally misclassified noise or artifacts as buildings.

D. Class Imbalance

The dataset was heavily imbalanced, with only 6.5% of pixels, on average, belonging to the building class, and a median building pixel proportion of just 3.4%. Additionally, approximately 10% of the image tiles contained no buildings at all, making it more difficult for the model to generalize effectively.

E. Key Techniques

To address the project challenges, we implemented three core strategies:

- **Pretrained Encoder:** Used an `xresnet34` backbone pretrained on ImageNet to benefit from transfer learning
- **Class Balancing:** Applied weighted cross-entropy loss (0.25 for background, 0.75 for buildings) to handle imbalanced data
- **Robust Training:** Enhanced the dataset with:
 - Random flips (horizontal/vertical)
 - 180° rotations
 - Brightness/contrast adjustments

F. Model Comparisons

While we didn’t test fundamentally different architectures, we evaluated several variations:

TABLE II: Model Variants Tested

Component	Option Tested	Outcome
Encoder	xresnet50	Rejected (6× slower, no improvement)
Activation	Mish	Rejected (unstable training)
Loss Function	Dice Loss	Rejected (poor convergence)
Architecture	+Self-Attention	Rejected (no significant gain)

G. Final Architecture

The selected model combines:

- **Encoder:** Pretrained `xresnet34` (optimal speed/accuracy balance)
- **Decoder:** Standard U-Net structure
- **Training:** 40 epochs with:
 - Batch size: 12
 - Learning rate: 3e-5 to 3e-4 (1-cycle policy)
 - Augmentations: Flips, rotations, lighting changes

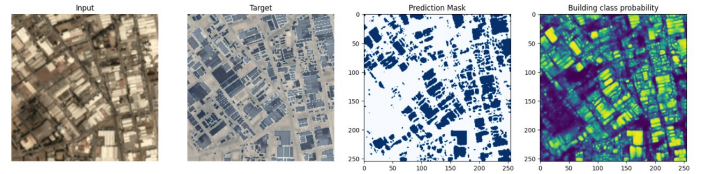


Fig. 3: target(2nd image) vs predictions (3 and 4th image)

H. Future Work

While the current pipeline achieved moderate success (Dice=0.58), several improvements are planned:

- **Architecture Enhancements:**
 - Overlapping tiles (50% stride) to better capture building boundaries
 - Temporal sequence modeling (LSTM/3D-CNN) to utilize multi-temporal data
- **Model Optimization:**
 - Systematic hyperparameter tuning (learning rate, batch size, augmentation intensity)

- Experimentation with deeper backbones (xresnet50, EfficientNet) upon GPU availability
- **Deployment Pipeline:**
 - Web API deployment using FastAPI when target Dice > 0.65 is achieved
 - Real-time prediction service with Azure ML Studio
 - Continuous learning system for model updates

VI. CONCLUSION

In this project, we addressed the task of building footprint segmentation from high-resolution satellite imagery using a subset of the SpaceNet 7 dataset. The primary objective was to classify each pixel into either Land or Building categories, a task made challenging by the severe class imbalance and variability inherent in real-world satellite data.

We employed a U-Net architecture with a pretrained xresnet34 encoder to perform binary segmentation. Several techniques, including weighted loss functions and data augmentation, were used to mitigate class imbalance and environmental variability. Through iterative experimentation, we optimized the training process and achieved a Dice Score of 0.58, indicating a moderate level of success in detecting small, isolated buildings under varying seasonal conditions.

While the model performed well in sparsely populated and rural areas, it faced difficulties in dense urban environments, where adjacent buildings were often merged, and large structures were occasionally missed. Moreover, false positives were noted in empty regions, highlighting areas for future improvement.

Although no systematic model benchmarking was conducted against alternative architectures such as DeepLabV3+ or Mask R-CNN, internal variations in encoder depth, activation functions, and loss functions provided valuable insights into the model's behavior.

In summary, this project demonstrates the potential of deep learning-based methods for building segmentation in satellite imagery while also highlighting the persistent challenges of class imbalance, urban complexity, and environmental noise. Future work could focus on more systematic architecture comparisons, enhanced loss functions (such as combined Dice and cross-entropy losses), and advanced augmentation strategies to further improve segmentation accuracy in challenging scenes.

REFERENCES

- [1] V. Iglovikov and A. Shvets, "TernausNet: U-Net with VGG11 encoder pre-trained on ImageNet for image segmentation," *arXiv preprint arXiv:1801.05746*, 2018.
- [2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2015, pp. 234–241.
- [3] A. Van Etten, D. Lindenbaum, and T. M. Bacastow, "SpaceNet: A Remote Sensing Dataset and Challenge Series," *arXiv preprint arXiv:1807.01232*, 2018.
- [4] A. Van Etten and D. Hogan, "SpaceNet 7 Multi-Temporal Urban Development Challenge," SpaceNet Challenge Documentation, 2020. [Online]. Available: <https://spacenet.ai/sn7-challenge/>
- [5] J. Howard and S. Gugger, *Deep Learning for Coders with fastai and PyTorch: AI Applications Without a PhD*. O'Reilly Media, 2020.
- [6] D. Misra, "Mish: A Self Regularized Non-Monotonic Activation Function," *arXiv preprint arXiv:1908.08681*, 2019.
- [7] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

SELF-DECLARATION FORM

The project was completed collaboratively with the following jointly completed tasks:

- Discussed and finalized the project topic.
- Selected the dataset together.
- Developed the project approach collectively.

The individual contributions are detailed below:

- **Manoj Kumar Surabhi** Responsible for code development.
- **Sumanth Reddy Thandra** Responsible for preprocessing of the code and preparation of the presentation slides.
- **Xiaolin Liu** Responsible for model selection and report writing.